



ROOT CAUSE UNLOCKED

Unlock Your *Hormones*

Your Hormones Are Talking. Here Is How to Hear What They Are Actually Saying, and What to Do About the Message.

A guide to the root causes behind PCOS, low testosterone, and HPA-driven hormonal symptoms, for women and men who have been told their labs are "normal" and still do not feel like themselves.

Dr. Adam Seay, DC

Root Cause Unlocked, Garner, NC · rootcauseunlocked.com · (919) 662-0520

Unlock Your Hormones

Your Hormones Are Talking. Here Is How to Hear What They Are Actually Saying, and What to Do About the Message.

BEFORE YOU START

A note before you start. This guide is educational information about how sex hormones, stress hormones, and metabolic function interact. It is not individualized medical advice, it does not diagnose any condition, and it does not replace a relationship with your physician or a clinical evaluation with my team.

A few things worth stating plainly before you read further:

This guide does not treat hormonal birth control, testosterone replacement therapy (TRT), or metformin as the enemy, and it never will. If you take hormonal birth control, understand that a regular-seeming cycle on the pill is a withdrawal bleed created by the medication itself, not necessarily proof that the underlying pattern this guide discusses has resolved. That is simply how the medication works. It is not a reason to stop or change your birth control. If you are already on prescribed TRT, or your physician is discussing it with you, nothing here should be read as a reason to start, stop, or adjust that therapy on your own. If you have been prescribed metformin for insulin resistance or PCOS, keep taking it exactly as prescribed. The nutrient and lifestyle strategies in this guide are additive to these treatments, not a substitute for any of them, and any change to a prescription is a conversation with the physician who prescribed it, not a decision to make from an ebook.

A note on fertility. Many people reading this because of a PCOS diagnosis, or a partner's diagnosis, are also trying to conceive, and that makes this subject emotionally loaded in a way that deserves honesty up front. This guide is not fertility treatment. Nothing here promises a pregnancy outcome. Cycle regularity and ovulation are discussed in these pages as physiology, information about how your body is functioning, because that is genuinely useful to understand. If conceiving is your goal, that conversation belongs with your physician or a reproductive endocrinologist, who can evaluate your specific situation properly. Please keep taking any prescribed medication exactly as directed, and talk to your doctor or a qualified practitioner before starting any new supplement, especially if you are pregnant, breastfeeding, or take other prescriptions.

Contents

INTRODUCTION

The Labs Came Back Normal. You Still Do Not Feel Like Yourself.

KEY 1

Your Hormones Are Messengers, Not the Root Cause

KEY 2

Insulin Resistance Is the Engine Behind Most PCOS

KEY 3

The HPA Axis and Your Sex Hormones Under Chronic Stress

KEY 4

Estrogen Metabolism, Detoxification, and the Estrobolome

KEY 5

Androgens Are Not Just “The Male Hormones”

KEY 6

What Is Actually Driving Low Testosterone in Men Under Fifty

KEY 7

Thyroid, the Cross-Referenced Driver

KEY 8

Testing Honestly: What Serum, Salivary, and Urine Hormone Testing Can and Cannot Tell You

Red Flags: When to Seek Care Now, Not Later

The Staged Roadmap

Cycle and Symptom Tracking Worksheet

Take the Next Step: Your Free Root Discovery Session

References

The Labs Came Back Normal. You Still Do Not Feel Like Yourself.

Somewhere between a normal TSH, a normal fasting glucose, and a doctor's visit that lasted twelve minutes, a lot of people stop asking questions about their hormones. The cycle is irregular but "not a big deal." The energy is gone but "that's just stress." The weight will not move no matter what changes. The libido quietly disappeared sometime in the last few years. For men, the drive and the muscle and the morning energy that used to be there without effort are just gone, and everyone around them says that is what happens after thirty-five.

None of that is a diagnosis. All of it is information.

Hormones are not the root cause of how you feel. They are the messengers. Estrogen, progesterone, testosterone, cortisol, and DHEA are downstream of what is actually happening in your body: how your cells respond to insulin, how loud your stress response has been running, how efficiently your liver and gut clear used hormones out of circulation, how well you sleep, and what your nutrient status actually is. Treating a hormone number in isolation, without asking what is driving it, is how people end up on a prescription that manages a symptom without ever touching the mechanism underneath it. That is not a criticism of the prescription. It is a case for looking one level deeper.

This guide covers three of the most common patterns I see walk through the door: polycystic ovary syndrome (PCOS), low testosterone in men under fifty, and the HPA axis, the stress system, and where it intersects sex hormones. It also covers what a hormone test can and cannot actually tell you, because that turns out to matter more than most people realize.

Below are eight keys. Each one explains a specific mechanism, in plain language, and what you can do about it at an educational level. The goal is not to hand you a diagnosis. It is to make sure that the next conversation you have with a practitioner, whether that is my team or your own physician, is a more informed one.

Your Hormones Are Messengers, Not the Root Cause

The question almost never asked. Most hormone conversations start and end with a single number: your testosterone, your estrogen, your TSH. If the number is in range, the conversation is over. If it is out of range, the fix is usually to replace or suppress that one hormone directly. What almost never gets asked is the more useful question: what is upstream of this number, and is it likely to keep happening again unless that upstream driver changes.

What actually drives the number. Every steroid hormone your body makes, cortisol, DHEA, testosterone, estrogen, and progesterone, starts from the same raw material: cholesterol, converted step by step through a shared enzyme cascade. That cascade does not run in a vacuum. It is shaped by how much inflammation your body is managing, how insulin-sensitive your tissues are, how well your liver clears hormones after they have done their job, what your gut microbiome is doing to hormones in circulation, and how loudly your nervous system perceives ongoing threat or stress. Change any one of those terrain factors and the downstream hormone picture moves, often without anyone touching the hormone itself. This is exactly why two people with an identical low testosterone reading, or an identical PCOS diagnosis, can have completely different underlying drivers, and why a plan that only targets the hormone number tends to help for a while and then plateau.

Where this guide goes from here. The rest of this guide walks through the terrain factors that most commonly drive the three patterns above: insulin resistance, chronic stress and the HPA axis, and estrogen and androgen metabolism through the liver and gut. None of it replaces a hormone panel. All of it is meant to make that panel make more sense.

What you are buying here, stated plainly. The nutrients discussed in this guide sit at different evidence tiers, and I grade each one where it appears rather than presenting them as a single uniform block. Ashwagandha has randomized trial evidence behind it. Inositol has trial evidence that the guideline reviewers themselves rated limited and inconclusive. DIM has one long randomized trial measuring a metabolite ratio rather than a symptom outcome. Most of the rest is laboratory work and reasoned clinical practice, which is what happens to compounds nobody can patent: the large outcome trials never get funded, so absence of evidence here is usually absence of a sponsor rather than absence of an effect. Two things follow, and both are worth acting on. Put your money into the terrain work first, sleep, resistance training, blood sugar, and the deficiencies you can actually measure, because that is where the evidence is strongest and it is also the cheapest part of the plan. Then hold anything you add on top of that to a defined endpoint rather than running it indefinitely.

Insulin Resistance Is the Engine Behind Most PCOS

The part the ovary framing misses. PCOS is usually described to patients as an ovarian problem: cysts on the ovaries, irregular periods, and acne or excess hair growth. That description misses the driver behind most of it. PCOS affects roughly 12 percent of women worldwide by the most common diagnostic standard, the Rotterdam criteria, making it one of the most common hormonal conditions in medicine, and it is closely tied to metabolic function, not just the ovaries.¹ A comprehensive 2024 review described PCOS as marked fundamentally by insulin resistance and hyperandrogenism together, a combination that meaningfully raises the risk of early-onset type 2 diabetes and cardiovascular disease over time, alongside the reproductive symptoms most people associate with the diagnosis.²

How insulin drives the pattern. Insulin does two things that directly drive the PCOS pattern. First, at high circulating levels, insulin acts on the ovary's theca cells and amplifies the effect of luteinizing hormone (LH), pushing those cells to produce more androgens (testosterone and its relatives) than they otherwise would. Second, insulin suppresses the liver's production of sex hormone binding globulin (SHBG), the protein that binds up circulating testosterone and keeps it inactive. Less SHBG means more free, active androgen reaching tissue, even without the ovary making more of it. Put those two effects together and you get the hallmark PCOS picture: irregular or absent ovulation, acne, excess facial or body hair growth, and, in many patients, difficulty losing weight, all downstream of insulin resistance rather than a primary ovarian defect.

The highest-leverage work. Because insulin resistance is the driver, addressing it is where the highest-leverage work happens: resistance training and regular movement (both directly improve cellular insulin sensitivity), reducing refined carbohydrate and added sugar intake, adequate protein and fiber at meals to blunt glucose swings, and correcting the nutrient deficiencies (magnesium and vitamin D among them) that commonly track alongside insulin resistance. Myo-inositol and D-chiro-inositol, nutrients naturally present in the body that support insulin signaling, are commonly discussed for PCOS, and it is worth being honest about what the evidence actually shows rather than what the marketing claims. A 2024 systematic review conducted specifically to inform the international PCOS treatment guidelines found real benefit for some metabolic measures and a potential benefit from D-chiro-inositol for ovulation, but rated the overall evidence limited and inconclusive, not the slam-dunk result inositol is often sold as.³ If you have been prescribed metformin for insulin resistance or PCOS, keep taking it exactly as prescribed. The strategies above are additive to metformin, not a substitute for it, and any change to that prescription belongs with the physician who prescribed it.

The HPA Axis and Your Sex Hormones Under Chronic Stress

A term that isn't a diagnosis. “Adrenal fatigue” is one of the most-searched terms in functional health, and it is not a recognized medical diagnosis. A 2016 systematic review that specifically set out to test whether “adrenal fatigue” was a real, substantiated condition reviewed 58 qualifying studies and concluded there is no consistent evidence for it as an actual medical entity, in large part because the cortisol-testing methods used to diagnose it produce conflicting, inconsistent results across the literature.⁴ That does not mean chronic stress has no effect on your hormones. It means the popular framing overstates a real, more nuanced mechanism, and the honest name for what is actually happening is HPA axis dysregulation, a disruption in the communication loop between your brain and your stress hormone system, not a pair of adrenal glands that have run out of gas.

What's actually happening between the two systems. Your hypothalamic-pituitary-adrenal (HPA) axis, the stress system, and your hypothalamic-pituitary-gonadal (HPG) axis, the reproductive hormone system, share upstream signaling and compete for some of the same downstream resources. Under sustained psychological stress, disordered eating patterns, or excessive exercise without adequate recovery, the brain can disrupt the pulsatile release of gonadotropin-releasing hormone (GnRH), which in turn suppresses the downstream signals that drive ovulation. This is well documented in a condition called functional hypothalamic amenorrhea, where cycles stop or become irregular under sustained physical or psychological stress load, with real downstream effects on bone health, mood, and cardiovascular risk if it continues unaddressed.⁵ A related idea you may have heard is “pregnenolone steal,” the theory that chronic stress shunts the shared hormone precursor pregnenolone toward cortisol production at the expense of DHEA and sex hormones. It is a biochemically plausible model, and it is worth being honest that direct human evidence for this specific substrate competition is limited. What is better supported is the broader pattern: the ratio between cortisol and DHEA, sometimes called the cortisol-to-DHEA ratio, is a recognized marker of how your stress system is functioning, and a ratio that skews toward more cortisol and less DHEA is associated with worse outcomes across aging, pain sensitivity, and frailty.⁶ Chronic stress also directly suppresses testosterone production in men, independent of any effect on cortisol.

What actually helps this pattern. Sleep, nervous system regulation, and addressing the actual sources of chronic stress do more for this pattern than any single supplement. Among the more studied adaptogenic herbs, ashwagandha has real, recent evidence behind it: a 2026 meta-analysis of 23 randomized controlled trials found it meaningfully lowered cortisol and, specifically in men, raised testosterone, with no comparable testosterone effect seen in women.⁷ That sex-specific difference matters and is exactly the kind of nuance worth knowing before assuming a supplement works the same way for everyone. This chapter intentionally stays brief on cortisol itself. The full mechanism of chronic stress, cortisol, and energy production is a subject large enough to deserve its own treatment, and I have kept it out of this one deliberately. What matters here is

the intersection: chronic, unaddressed stress is a legitimate upstream driver of cycle irregularity in women and lower testosterone in men, deserving real attention, without needing to invoke a diagnosis that does not exist to take it seriously.

Estrogen Metabolism, Detoxification, and the Estrobolome

The word almost everyone misuses. Estrogen dominance is a term you have probably heard, usually attached to symptoms like heavy or painful periods, breast tenderness, mood swings, and stubborn weight around the hips and thighs. What gets left out of most explanations is that estrogen dominance is rarely about making too much estrogen. It is much more often about clearing it too slowly, and that clearance happens in two places most people never think about: the liver and the gut.

How clearance actually works. After estrogen has done its job, your liver processes it in two phases. Phase 1 hydroxylates the estrogen molecule down one of several pathways, most notably the 2-hydroxy pathway (generally considered the more favorable route) versus the 16-alpha-hydroxy pathway (a more potent, longer-acting route). Phase 2 then attaches a molecule (methylation, via the COMT enzyme, or glucuronidation) that tags the processed estrogen for elimination. Once tagged for elimination, it travels to the gut to be excreted. Here is the part almost nobody explains: certain gut bacteria produce an enzyme called beta-glucuronidase, which can strip that elimination tag back off, effectively re-releasing already-processed estrogen back into circulation instead of letting it leave the body. This gut-microbiome effect on estrogen has a name in the research literature, the estrobolome, and a disrupted, less diverse gut microbiome with elevated beta-glucuronidase activity is a documented contributor to estrogen-related symptoms and disease risk.⁸ Xenoestrogens, estrogen-mimicking compounds from plastics (BPA), certain personal care products (parabens), and some pesticides, add to the total estrogenic load your liver and gut have to clear, which is why reducing exposure to them is a legitimate, low-risk first step regardless of what your labs show.

Where the leverage actually is. Supporting healthy Phase 1 to Phase 2 liver clearance and a healthy gut microbiome are the two levers that matter most here, alongside reducing xenoestrogen exposure where it is easy to do so. Cruciferous vegetables (broccoli, cauliflower, Brussels sprouts) contain compounds that support the more favorable 2-hydroxy estrogen pathway, and DIM (diindolylmethane), a concentrated compound derived from those same vegetables, has real human trial evidence behind it: a randomized, placebo-controlled trial found that daily DIM shifted the urinary estrogen metabolite ratio toward the more favorable pathway and raised SHBG, compared to placebo, over twelve months.⁹ That same trial is also a genuine safety reminder rather than just a win: it was conducted in women taking tamoxifen, a prescription anti-estrogen medication, and DIM measurably reduced the levels of active tamoxifen metabolites in their blood. If you are on tamoxifen, another prescribed hormone therapy, or any medication that affects estrogen metabolism, that is a conversation for your prescriber before adding DIM or any similar compound, not a decision to make on your own. Calcium-D-glucarate, a compound that inhibits the same beta-glucuronidase enzyme responsible for re-releasing estrogen in the gut, is the other commonly discussed nutrient in this category. As for how much this actually matters for long-term risk, it is worth being precise rather than alarmist: a case-control study of the 2-hydroxy to 16-alpha-hydroxy estrogen metabolite ratio found a significant association with breast cancer risk in postmenopausal women, but not in premenopausal women,

where naturally fluctuating hormone levels across the cycle appear to make the ratio less meaningful as a standalone marker.¹⁰ That is a real finding, and it is also a narrower one than the way this topic often gets marketed.

A note on pregnancy and breastfeeding. Several of the compounds discussed in this guide, including DIM, calcium-D-glucarate, berberine, and ashwagandha, are not recommended during pregnancy or while breastfeeding. If you are pregnant, trying to conceive, or nursing, talk to your physician before starting any of them.

Androgens Are Not Just “The Male Hormones”

Two blind spots from one assumption. Testosterone gets treated culturally as the male hormone and estrogen as the female one, which leads to two blind spots: women’s androgen levels get almost no clinical attention unless they are too high (as in PCOS), and men’s estrogen, produced in meaningful amounts via a process called aromatization, gets almost none at all. Both sexes make, need, and are affected by the full spectrum of sex hormones.

What androgens actually do for each sex. Androgens, testosterone, DHEA, androstenedione, and DHT among them, play a real role in women’s health beyond reproduction: bone density and remodeling, cardiovascular health, muscle mass and tone, and cognitive function all depend in part on adequate androgen levels, not just adequate estrogen.¹¹ Low androgen states in women, whether from surgical removal of the ovaries, adrenal insufficiency, or the natural decline in DHEA-S with age, are a real and underappreciated pattern, distinct from the high-androgen picture seen in PCOS. In men, the enzyme aromatase converts a portion of circulating testosterone into estradiol, and adipose (fat) tissue is a major site of that conversion, meaning body composition directly affects a man’s estrogen-to-testosterone balance, not just his testosterone number in isolation.

What this means for you, by sex. For women, the practical takeaway is that a “normal” estrogen and progesterone panel does not tell the whole story, and androgen status (total and free testosterone, DHEA-S) is worth including in a complete hormone workup, particularly if energy, libido, or muscle mass are part of the symptom picture. For men, body composition, addressed in the next key, is directly relevant to androgen balance in both directions, not just the testosterone number. Any decision to supplement DHEA or pursue hormone therapy of any kind, for either sex, is a conversation with a practitioner working from your actual labs, not a decision to make from general information in a guide like this one.

What Is Actually Driving Low Testosterone in Men Under Fifty

The demographic shift nobody predicted. Low testosterone used to be framed as an older man's condition. It increasingly is not. A population-level study tracking American men across multiple survey years found that testosterone levels have declined across birth cohorts independent of age itself, a decline not explained by the usual suspects like rising obesity or smoking rates.¹² Something in modern life, beyond simple aging, is lowering testosterone in younger men, and the good news is that several of the biggest drivers are genuinely modifiable.

The biggest modifiable levers. Sleep is the most underrated lever here. In a controlled study, healthy young men restricted to five hours of sleep per night for one week saw daytime testosterone levels drop by 10 to 15 percent, a decline larger than the 1 to 2 percent per year testosterone naturally falls with normal aging.¹³ Body composition is the second major lever: obesity is the clinical condition most strongly associated with lowered testosterone in men, working through two separate mechanisms depending on severity, reduced SHBG in modest excess weight, and genuine suppression of the brain-testicular signaling axis in more significant obesity, driven partly by inflammatory signaling and partly by the aromatase conversion of testosterone to estrogen in fat tissue described in the previous key.¹⁴ Blood sugar dysregulation compounds both of the above, since insulin resistance independently suppresses SHBG in men, the same mechanism that raises free androgens in women with PCOS, just running in the same biochemical direction toward a different clinical picture. Chronic stress adds a fourth layer, suppressing testosterone production through the same HPA-HPG crosstalk covered in Key 3. Nutrient status matters too, though it is worth being precise about which nutrients actually have evidence behind them and which are oversold. A 2025 study of young men found several trace elements, including zinc, iron, and sulfur, associated with more favorable testosterone levels, while others, including phosphorus and strontium, were associated with lower levels.¹⁵ Vitamin D is a more honest story than most wellness content admits: the most rigorous recent meta-analysis of randomized controlled trials, published in 2026, found no clear effect of vitamin D supplementation on total testosterone or related markers in adult men, despite how often vitamin D is marketed as a testosterone booster.¹⁶ Correcting a genuine vitamin D deficiency still matters for bone, immune, and overall metabolic health. It is just not, on the current evidence, a reliable lever for testosterone specifically.

Where to focus first. In order of leverage: prioritize seven to nine hours of consistent sleep, build and maintain lean muscle through resistance training while addressing excess body fat, stabilize blood sugar through the same strategies covered in Key 2, and manage chronic stress directly rather than assuming it will resolve on its own. Ashwagandha, covered in Key 3, is one of the few supplements with real randomized-trial evidence for raising testosterone specifically in men.⁷ If you are already on prescribed testosterone replacement therapy (TRT), or your physician has raised it as an option, nothing in this guide is a reason to start, stop, or adjust that treatment yourself. Low testosterone that is confirmed on repeat morning

bloodwork, alongside real symptoms, is a legitimate medical finding that deserves a proper workup with your physician, not a self-diagnosis from a symptom list, and whether testosterone therapy is appropriate for your specific case is that physician's call to make with you.

Thyroid, the Cross-Referenced Driver

Where thyroid and sex hormones overlap. Thyroid function and sex hormone function are wired together more closely than most people realize, which means an unaddressed thyroid issue can look, on the surface, exactly like a primary sex hormone problem.

How the two systems are wired together. Thyroid hormone status influences the menstrual cycle both directly, through effects on the ovary, and indirectly, through its effect on sex hormone binding globulin (SHBG), prolactin, and gonadotropin-releasing hormone (GnRH) signaling.¹⁷ That means an underactive or autoimmune thyroid can drive irregular cycles, low libido, and fatigue that look identical to a primary hormonal pattern, and a hormone panel alone will not distinguish between the two. Hashimoto's thyroiditis, an autoimmune condition and the most common cause of hypothyroidism, is common enough among people already dealing with PCOS and other hormonal patterns that it is always worth ruling in or out, not assumed away because a single TSH came back "normal."

What's worth pursuing before assuming. If thyroid symptoms (cold intolerance, hair thinning, unexplained weight change, constipation, or persistent fatigue despite a full sleep and stress correction) are part of your picture, a full thyroid panel, not just TSH, is worth pursuing before assuming the pattern is purely a sex hormone issue. The complete framework for autoimmune thyroid disease, its triggers, its testing, and its treatment sequence lives in the companion guide, *Unlock Your Hashimoto's*. This guide will not re-cover that ground. The short version relevant here is simply this: do not let a "normal" TSH close the door on thyroid as a contributing driver of your hormonal symptoms.

Testing Honestly: What Serum, Salivary, and Urine Hormone Testing Can and Cannot Tell You

The differences that actually matter. Not all hormone tests measure the same thing, and the differences matter more than most patients are told. Choosing the wrong sample type for the question you are actually asking is one of the most common, and most avoidable, sources of confusing or contradictory hormone results.

What each sample type actually shows. Serum (blood) testing measures total hormone levels, bound and unbound, at a single point in time. It is the current standard of care for monitoring hormone replacement therapy and is the only sample type that reliably captures SHBG alongside your sex hormones, but a single blood draw cannot show you a cortisol rhythm across the day, and it misses most free, unbound hormone and metabolite information entirely. Urine testing captures hormone metabolites, meaning it can show you how your body is actually processing hormones (which liver pathway your estrogen is taking, for example), not just how much hormone is circulating, but interpretation requires care: a low result can reflect either low production or increased tissue uptake, and a high result can reflect either overproduction or a backup in clearance. Saliva testing reflects the free, bioavailable fraction of a hormone rather than the bound majority, which makes it genuinely useful for tracking a cortisol rhythm across the day, but it does not measure hormone metabolites, and, as the “adrenal fatigue” research above makes clear, the cortisol-testing literature built on salivary methods has produced inconsistent, sometimes conflicting results across studies, which is a real limitation worth naming rather than glossing over.⁴ None of these three approaches is universally “better.” Each answers a different question, and the right choice depends on what you are actually trying to find out.

Matching the test to the question. If you are being screened for a first-pass hormonal workup, serum testing covering your core sex hormones, SHBG, and a full thyroid panel is a reasonable, well-validated starting point. If cycle-related patterns or estrogen metabolism specifically are the question, a urine-based hormone metabolite panel adds real information a single blood draw cannot. If cortisol rhythm across the day is the specific question, salivary or urine testing done properly, at the right times of day, across a full cycle if you are still menstruating, adds value that a single morning blood cortisol cannot. What no single test does, regardless of sample type, is diagnose you on its own. Testing informs a clinical picture. It does not replace one.

Red Flags: When to Seek Care Now, Not Later

This guide is educational and does not diagnose. Some symptoms warrant a prompt conventional medical evaluation rather than a supplement or a wait-and-see approach, regardless of what else is going on hormonally:

- **Postmenopausal bleeding** (any vaginal bleeding after menopause) always warrants prompt evaluation.
- **A rapidly virilizing presentation** (fast-developing deepening of the voice, significant new hair growth, or clitoral enlargement over weeks to months, rather than a gradual pattern over years) can signal an androgen-secreting tumor and needs prompt workup.
- **Severe, acute pelvic pain**, especially sudden in onset, needs same-day evaluation.
- **A new testicular mass or lump** always warrants prompt evaluation and should never be watched and waited on.
- **Galactorrhea** (spontaneous milk-like nipple discharge unrelated to pregnancy or nursing) can indicate a pituitary issue and warrants a prolactin level and follow-up.
- **Adrenal insufficiency (Addison's disease) and pituitary disease** are real, distinct medical conditions that can present with fatigue and low hormone levels that look similar to the patterns in this guide. They require conventional diagnostic workup, and this guide is not a substitute for ruling them out if your presentation or labs suggest either one.

None of the above are things to address with supplements or lifestyle change first. See a physician promptly.

The Staged Roadmap

Hormonal patterns do not resolve on a fixed calendar, and healing is rarely a straight line. The general pace below reflects realistic timelines for the mechanisms covered in this guide, not a guarantee for any individual case.

Weeks 1 to 4: Foundation. Sleep consistency, blood sugar stabilization (protein and fiber at meals, reduced refined carbohydrate), reducing xenoestrogen exposure, and beginning to address the biggest single stress load in your life. Early energy and sleep improvements are typically the first thing people notice in this window.

Months 2 to 3: Cycle and pattern changes. For women with PCOS-driven irregularity, this is typically when cycle-length changes become noticeable, often across two to three cycles, as insulin sensitivity begins to shift. For men, this is when body composition changes from consistent resistance training and sleep begin to show up in energy and libido.

Months 3 to 6: Metabolic and hormonal consolidation. This is the window where insulin sensitivity, estrogen clearance patterns, and cortisol rhythm changes typically stabilize into a new, more consistent baseline, assuming the foundational work from the first two phases has been consistent.

Months 6 to 12: Full pattern stability. Full hormonal rebalancing, for both the PCOS and low-testosterone patterns covered here, typically takes six to twelve months of consistent terrain work to reach a stable, durable baseline, longer in more severe or longstanding cases. This is also the point where repeat lab testing is most useful for confirming what your body has already started telling you symptomatically.

If you are not seeing meaningful movement by the three-month mark, that is useful information too. It usually means an upstream driver has not yet been identified, not that the approach has failed.

Cycle and Symptom Tracking Worksheet

Whether you are tracking a menstrual cycle or simply tracking symptoms day to day, consistent tracking is one of the most useful, lowest-cost tools available, both for your own awareness and for a more productive conversation with any practitioner you see. Copy this table (paper, spreadsheet, or notes app) and fill it in daily.

Date	Cycle day (or N/A)	Sleep (hours, quality 1 to 5)	Energy (1 to 5)	Mood (1 to 5)	Stress (1 to 5)	Libido (1 to 5)	Skin, hair, or weight changes	Other symptoms	Notes

Monthly reflection questions: 1. Did cycle length, regularity, or flow change this month compared to last month (if applicable)? 2. What was your average sleep quality, and did anything specific improve or disrupt it? 3. What was the single largest stress load this month, and how did you handle it? 4. Did energy and mood track together, or did one lag behind the other? 5. What changed in your routine this month (diet, training, sleep, supplements, medications) that might explain any shifts above?

Bring this log to your next Root Discovery Session or physician visit. A month of consistent tracking tells a practitioner more than a single lab draw ever can on its own.

Take the Next Step: Your Free Root Discovery Session

The Root Discovery Session is a free consultation to review your specific situation: your symptom history, what testing (if any) you have already had, and what a reasonable next step looks like for your case. There is no obligation, and if you decide to move forward, this conversation is simply the first step.

Schedule your free Root Discovery Session at rootcauseunlocked.com/discovery.

You can also reach the office directly at (919) 662-0520.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

If you would rather start on your own

If you would rather start on your own, start by tracking. The cycle and symptom tracking worksheet earlier in this guide works on paper. The digital version does the same job on your phone, keeps your history in one place, and shows you what a single lab draw cannot: how symptoms move across a cycle or across a month, which is exactly the context that makes hormone testing interpretable instead of a snapshot. Two weeks of honest data makes any future appointment, with our team or your own physician, dramatically more useful than memory alone.

Start your free tracker at rootcauseunlocked.com/tracker.

Seay Wellness | 503 US-70 East, Suite L, Garner, NC 27529
rootcauseunlocked.com

This guide is provided for general educational purposes and does not constitute individualized medical advice, diagnosis, or treatment, and it is not fertility treatment. Always consult your physician before starting any new supplement or exercise program, particularly if you are pregnant, breastfeeding, or taking prescription medications, including hormonal birth control, testosterone replacement therapy, or metformin. Continue any prescribed medication exactly as directed by your physician.

References

Every citation below was independently retrieved and confirmed against its live PubMed record (eSearch plus eSummary and/or eFetch, including full-text verification via PubMed Central where the indexed abstract was incomplete) before use. Full verification detail, including claims investigated and left uncited, is in the companion file Hormones-Citation-Ledger.md.

1. Neven ACH, Forslund M, Ranasinha S, Sethi P, Dhungana RR, Mousa A, Tay CT, Teede H, Boyle JA. Prevalence of polycystic ovary syndrome: a global and regional systematic review and meta-analysis. *Human Reproduction Update*. 2026 May 1;32(3):277-312. PMID 41528735. <https://pubmed.ncbi.nlm.nih.gov/41528735/>
2. Stener-Victorin E, Teede H, Norman RJ, Legro R, Goodarzi MO, Dokras A, Laven J, Hoeger K, Piltonen TT. Polycystic ovary syndrome. *Nature Reviews Disease Primers*. 2024 Apr 18;10(1):27. PMID 38637590. <https://pubmed.ncbi.nlm.nih.gov/38637590/>
3. Fitz V, Graca S, Mahalingaiah S, Liu J, Lai L, Butt A, Armour M, Rao V, Naidoo D, Maunder A, Yang G, Vaddiparthi V, Witchel SF, Pena A, Spritzer PM, Li R, Tay C, Mousa A, Teede H, Ee C. Inositol for Polycystic Ovary Syndrome: A Systematic Review and Meta-analysis to Inform the 2023 Update of the International Evidence-based PCOS Guidelines. *Journal of Clinical Endocrinology and Metabolism*. 2024 May 17;109(6):1630-1655. PMID 38163998. <https://pubmed.ncbi.nlm.nih.gov/38163998/>
4. Cadegiani FA, Kater CE. Adrenal fatigue does not exist: a systematic review. *BMC Endocrine Disorders*. 2016 Aug 24;16(1):48. PMID 27557747. <https://pubmed.ncbi.nlm.nih.gov/27557747/>
5. Saadedine M, Kapoor E, Shufelt C. Functional Hypothalamic Amenorrhea: Recognition and Management of a Challenging Diagnosis. *Mayo Clinic Proceedings*. 2023 Sep;98(9):1376-1385. PMID 37661145. <https://pubmed.ncbi.nlm.nih.gov/37661145/>
6. Erceg N, Micic M, Forouzan E, Knezevic NN. The Role of Cortisol and Dehydroepiandrosterone in Obesity, Pain, and Aging. *Diseases*. 2025 Feb 1;13(2):42. PMID 39997049. <https://pubmed.ncbi.nlm.nih.gov/39997049/>
7. Fornalik M, Malkiewicz A, Adamczak D, Nawrocki F, Zielinska A, Mondal SA. Hormonal Modulation with Withania somnifera: Systematic Review and Meta-Analysis of Randomized-controlled Trials. *Planta Medica*. 2026 Jul;92(8):790-805. PMID 41740946. <https://pubmed.ncbi.nlm.nih.gov/41740946/>
8. Hu S, Ding Q, Zhang W, Kang M, Ma J, Zhao L. Gut microbial beta-glucuronidase: a vital regulator in female estrogen metabolism. *Gut Microbes*. 2023 Jan-Dec;15(1):2236749. PMID 37559394. <https://pubmed.ncbi.nlm.nih.gov/37559394/>
9. Thomson CA, Chow HHS, Wertheim BC, Roe DJ, Stopeck A, Maskarinec G, Altbach M, Chalasani P, Huang C, Strom MB, Galons JP, Thompson PA. A randomized, placebo-controlled trial of diindolylmethane for breast cancer biomarker modulation in patients taking tamoxifen. *Breast Cancer Research and Treatment*. 2017 Aug;165(1):97-107. PMID 28560655. <https://pubmed.ncbi.nlm.nih.gov/28560655/>
10. Ruan X, Seeger H, Wallwiener D, Huober J, Mueck AO. The ratio of the estradiol metabolites 2-hydroxyestrone (2-OHE1) and 16alpha-hydroxyestrone (16-OHE1) may predict breast cancer risk in postmenopausal but not in premenopausal women: two case-control studies. *Archives of Gynecology and Obstetrics*. 2015 May;291(5):1141-6. PMID 25318606. <https://pubmed.ncbi.nlm.nih.gov/25318606/>
11. Bianchi VE, Bresciani E, Meanti R, Rizzi L, Omeljaniuk RJ, Torsello A. The role of androgens in women's health and wellbeing. *Pharmacological Research*. 2021 Sep;171:105758. PMID 34242799. <https://pubmed.ncbi.nlm.nih.gov/34242799/>

12. Travison TG, Araujo AB, O'Donnell AB, Kupelian V, McKinlay JB. A population-level decline in serum testosterone levels in American men. *Journal of Clinical Endocrinology and Metabolism*. 2007 Jan;92(1):196-202. PMID 17062768. <https://pubmed.ncbi.nlm.nih.gov/17062768/>
13. Leproult R, Van Cauter E. Effect of 1 week of sleep restriction on testosterone levels in young healthy men. *JAMA*. 2011 Jun 1;305(21):2173-4. PMID 21632481. <https://pubmed.ncbi.nlm.nih.gov/21632481/>
14. Grossmann M. Hypogonadism and male obesity: Focus on unresolved questions. *Clinical Endocrinology*. 2018 Jul;89(1):11-21. PMID 29683196. <https://pubmed.ncbi.nlm.nih.gov/29683196/>
15. Tanaka T, Kojo K, Suetomi T, Nagumo Y, Midorikawa H, Matsuda T, Nakazono A, Shimizu T, Fujimoto S, Ikeda A, Kandori S, Negoro H, Takayama T, Nishiyama H. Distinct Clusters of Testosterone Levels, Symptoms, and Serum Trace Elements in Young Men: A Cross-Sectional Analysis. *Nutrients*. 2025 Feb 28;17(5):867. PMID 40077734. <https://pubmed.ncbi.nlm.nih.gov/40077734/>
16. Paez-Allendes L, Valenzuela-Fuenzalida JJ, Moya MP, Oyanedel G, Cifuentes-Suazo G, Figueroa-Puig J, Orellana-Donoso M, Mateluna-Valls E, Cabezas-Salgado JJ, Sanchis-Gimeno J, Bruna-Mejias A. Vitamin D Supplementation, Total Testosterone, and Androgen Bioavailability Markers in Adult Men: A Systematic Review and Meta-Analysis of Randomized Controlled Trials. *Nutrients*. 2026 Jun 26;18(13):2090. PMID 42451097. <https://pubmed.ncbi.nlm.nih.gov/42451097/>
17. Poppe K, Velkeniers B, Glinioer D. Thyroid disease and female reproduction. *Clinical Endocrinology*. 2007 Mar;66(3):309-21. PMID 17302862. <https://pubmed.ncbi.nlm.nih.gov/17302862/>

For educational purposes only. This guide does not constitute medical advice, diagnosis, or treatment, and reading it does not establish a patient-provider relationship.

Root Health | Dr. Adam Seay, DC | Garner, NC | rootcauseunlocked.com